THOMAS V. THOMPSON II, PH.D.

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ACADEMIC DEGREES & HIGHER EDUCATION

Ph.D. in Computer Science / University of Utah, Salt Lake City, UT

- Dissertation: "Haptic Rendering of Trimmed NURBS Models Within an Active Prototyping Environment"
- Specialty Areas: Computer-Aided Geometric Design (CAGD), Computer Graphics, Subdivision Surfaces, Distributed Systems

M.S. in Computer Science / Wake Forest University, Winston-Salem, NC

- Thesis: "Three Dimensional Polygon-Mesh Morphology" (Minor in Radiology)

B.S. in Computer Science / West Virginia Institute of Technology, Montgomery, WV

- Honors: Alpha Chi Honor Fraternity, L.C. Nelson Presidential Leadership Award (Minor in Mathematics)

HIGHER EDUCATION INSTRUCTION & ACADEMIC EXPERIENCE

University of Utah / School of Computing | Salt Lake City, UT
Research Associate & Technical Lead

- Served as a Technical Lead for the Alpha_1 Geometric Modeling System research software group, mentoring graduate and undergraduate students in software architecture, C++ object-oriented design, and system integration.
- Provided ongoing academic and technical consultations to students during regular office hours, guiding coding practices and research implementations.
- Graduate Teaching Assistant & Discussion Leader
- Conducted weekly computer laboratory sessions, led discussion sections, and held office hours for Intro to Computer Science II, Intro to Computer Graphics, and Computer-Aided Geometric Modeling.
- Evaluated and graded student programming projects, lab assignments, and examinations, providing constructive feedback to support student learning outcomes.
- Delivered primary course lectures in place of lead faculty during absences.

Wake Forest University / Department of Mathematics & Computer Science | Winston-Salem, NC
Graduate Assistant & Laboratory Instructor

- Instructed hands-on computer laboratory sessions and assisted lead professors for Intro to Computer Science and Computer Graphics.
- Held office hours for undergraduate student tutoring, graded coding projects, and assisted students in debugging software.

TEACHING COMPETENCIES & CURRICULUM ALIGNMENT

- Introductory Programming (Python, C++, C): Fundamental logic, control structures, functions, memory management, and procedural programming.
- Data Structures & Algorithms: Arrays, linked lists, stacks, queues, trees, graphs, sorting, searching, and algorithmic complexity (O(n) notation).
- Object-Oriented Design & Software Engineering: Classes, inheritance, polymorphism, design patterns, modular architecture, and version control (Git).
- Computer Graphics & Procedural Geometry: 3D coordinate systems, linear algebra transformations, rendering pipelines, and shader concepts.
- Computer Organization & Assembly Language: Machine architecture, memory hierarchies, registers, and low-level system execution.

INDUSTRY ENGINEERING & TECHNICAL LEADERSHIP

Lightspeed LA / Irvine, CA Mar 2023 – Jul 2026
Senior Tools Engineer III

- Architected cinematic narrative pipeline tools and custom Unreal Engine 5 integrations.
- Authored hair/fur shader flow map conversion tools, streamlining artist pipelines.
- Conducted technical documentation, workflow training, and team onboarding for custom toolsets.

Imagendary Studios / Irvine, CA Feb 2022 – Jan 2023
Senior Core Software Engineer

- Led "Core Team" initiatives designing cinematic asset pipelines, DCC integrations, and Maya to Unreal Engine 5 export workflows.
- Created detailed technical walkthroughs and held training sessions for remote production teams.

Blizzard Entertainment / Irvine, CA Mar 2012 – Feb 2022
Senior Software Engineer II

- Architected end-to-end rendering and procedural geometry systems (Belaro, MobBoss, Scatter).
- Acted as primary technical liaison and educator for external software partners including Autodesk, Pixar, and Redshift.
- Mentored mid-level and junior engineers on software architecture, performance optimization, and procedural algorithms.

Walt Disney Animation Studios / Burbank, CA Jan 2001 – Mar 2012
Senior Development Software Engineer & Software Supervisor

- Lead architect of XGen (arbitrary instancing engine), now integrated into Autodesk Maya as a global industry standard.
- Three time Academy Scientific and Technical Award Nominee (Motion Picture Academy) - Recognized for the architectural creation and industry impact of XGen, iGroom, and Bonsai.
- Managed and supervised a tools engineering team of 20+ software developers and technical directors.
- Designed and executed new-developer onboarding bootcamps, structured guided "starter" tasks, and conducted technical code reviews.
- Developed and delivered multi-studio technical training workshops for artists and technical directors at Disney Animation, Pixar, ImageMovers Digital, Digital Domain, and Disney Toon Studios.

PEDAGOGICAL APPROACH & STUDENT DIVERSITY

- Experiential & Visual Pedagogy: Proven track record of translating complex abstract mathematics (e.g., guide interpolation algorithms, NURBS mathematics) into intuitive, interactive visual models and hands-on tutorials for non-technical learners.
- Inclusive Mentorship: Decades of experience onboarding and mentoring engineers, interns, and studio artists from diverse socio-economic, educational, and technical backgrounds.

SCHOLARLY PUBLICATIONS & CONFERENCE PRESENTATIONS

- Thompson II, T.V., "Determining Guide Connectivity," U.S. Patent 8,823,714, 2014.
- Thompson II, T.V., Jenkins, S., and Tappan, C., “Let There Be Hair,” Proc. Siggraph, (Vancouver), 2011.
- Thompson II, T.V., Petti, E.J., and Tappan, C., “XGen: Arbitrary Primitive Generator,” Proc. Siggraph, (San Diego, CA), 2003.
- Petti, E.J., Thompson II, T.V., Lusinsky, A., and Driskill, H., “Dragon Scales: The Evolution of ScaleTool for Reign of Fire,” Proc. Siggraph, (San Antonio, TX), 2002.
- Thompson II, T.V., and Cohen, E., "Haptic Methods for Rendering Trimmed NURBS Models in Real-Time," *Sixth ACM Symposium on Solid Modeling and Applications*, (Ann Arbor, MI), 2001.
- De St. Germain, H.J., and Thompson II, T.V., "Point Dispersion for RE Purposes," *Sixth ACM Symposium on Solid Modeling and Applications*, (Ann Arbor, MI), 2001.
- Hollerbach, J.M., Cohen, E., Thompson, W.B., Freier, R., Johnson, D.E., and Thompson II, T.V., "Virtual prototyping for human-centric design," *NSF Design and Manufacturing Research Conference*, (Vancouver), 2000.
- Thompson II, T.V., and Cohen, E., "Direct Haptic Rendering of Complex Trimmed NURBS Models," *Proc. 8th Annual Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems*, ASME, (Nashville, TN), 1999.
- Johnson, D.E., Thompson II, T.V., Kaplan, M., Nelson, D., and Cohen, E., "Painting Textures with a Haptic Interface," *Proc. Virtual Reality '99*, IEEE, (Houston, TX), 1999.
- Thompson II, T.V., Johnson, D.E., and Cohen, E., "Direct Haptic Rendering Of Sculptured Models," *Proc. Symposium on Interactive 3D Graphics (I3D)*, ACM, (Providence, RI), 1997.
- Hollerbach, J.M., Cohen, E., Thompson, W.B., Freier, R., Johnson, D., Nahvi, A., Nelson, D., Thompson II, T.V., and Jacobsen, S.C., "Haptic Interfacing For Virtual Prototyping Of Mechanical CAD Designs," *ASME Design for Manufacturing Symposium,* (Sacramento, CA), 1997.
- Thompson II, T.V., Nelson, D.D., Cohen, E., and Hollerbach, J.M., "Maneuverable Models Within A Haptic Virtual Environment," *Proc. 6th Annual Symposium on Haptic Interfaces*, ASME, (Dallas, TX), 1997.